



Analytics Solutions Bootcamp: #04

Data Analysis and Business Intelligence Essentials

An introduction on descriptive and diagnostic analytics and building scalable dashboards

Hello, everyone! Welcome to the fourth module of the Analytics Solutions Bootcamp. Today, you will learn about the essentials of data analysis and business intelligence.

Topics to be covered



- What is data analysis?
- Data analysis process
- Exploratory data analysis (EDA)
- Fundamentals of data visualization
- Descriptive analytics
- Diagnostic analytics
- Business intelligence primer

Defining data analysis and the data analysis process

Understand what data analysis is and how we perform data analysis

Defining data analysis and the data analysis process

Defining data analysis



Analytics is the process of collecting and analyzing data to discover, interpret, and communicate meaningful insights.

Using stents to prevent strokes >>

Group	30 days		365 days	
	Stroke	No event	Stroke	No event
Control	13	214	28	199
Treatment	33	191	45	179
Total	46	405	73	378

Case study: Descriptive statistics for the stent study. In the first 30 days of the study, 13 (5.72%) out of 227 patients in the control group had stroke, while 33 (14.73%) out of 224 patients in the treatment group had stroke. For the rest of the year, the patients who had stroke rose to 28 (12.33%) and 45 (20.09%) respective to control and treatment groups (Chimowitz et al. 2011).

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A stent is a tiny, expandable tube made of metal mesh, fabric, or silicone used to prop open narrow or blocked passageways in the body. They are most commonly placed in heart (coronary) arteries, but can also support blood vessels in the brain or legs, airways in the lungs, and urinary tubes (ureters).

Many doctors have hoped that there would be similar benefits for patients at risk of stroke. We start by writing the principal question the researchers hope to answer:

Does the use of stents reduce the risk of stroke?

The researchers who asked this question conducted an experiment with 451 at-risk patients. Each volunteer patient was randomly assigned to one of two groups:

Treatment group. Patients in the treatment group received a stent and medical management. The medical management included medications, management of risk factors, and help in lifestyle modification.

Control group. Patients in the control group received the same medical management as the treatment group, but they did not receive stents.

Conclusion:

The 2011 SAMMPRIS trial by Chimowitz et al. **concluded that aggressive medical management is superior to stenting with the Wingspan system for symptomatic intracranial arterial stenosis**, due to high 30-day stroke or death rates in the stenting group. Medical therapy achieved a 1-year primary endpoint rate of 12.2% compared to 20.0% in the stenting cohort. For more details, visit [NEJM](#). [1, 2, 3, 4, 5]

Defining data analysis and the data analysis process

Takeaway from the case study >>

Data analysis is a great instrument for many domains.

- Healthcare: clinical trials and drug development, public health and epidemiology
- Sports: player performance and game strategy
- Agriculture: precision farming, yield prediction, livestock monitoring
- Transportation and logistics: traffic management, ride sharing
- Environmental science: climate modeling, disaster response

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Healthcare: cite example on COVID pandemic

Sports: cite example from Young Sheldon

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Healthcare: cite example on COVID pandemic

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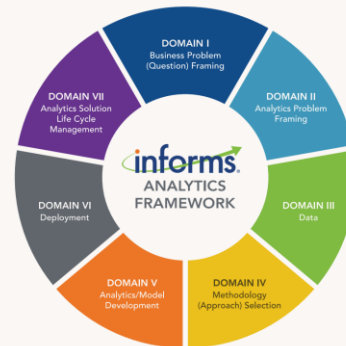
Defining data analysis and the data analysis process

The data analysis process



The INFORMS analytics framework shows the steps of doing data analytics projects.

We will use this framework throughout this module.



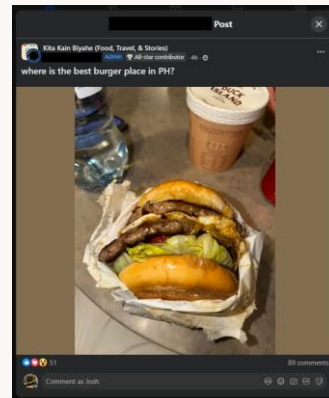
Defining data analysis and the data analysis process

I. Business problem framing



This includes activities related to understanding the business problem and relevant questions, as well as evaluating the scope of the problem.

Ex. "Where is the best burger place in PH?"



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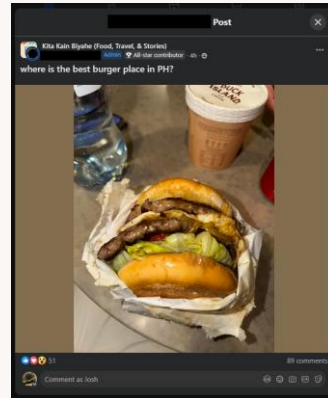
Defining data analysis and the data analysis process

II. Analytics problem framing



Translates a clearly defined business problem (question) into a structured analytics problem that can be addressed with analytics techniques applied to data.

Ex. "Which burger places have great customer ratings?"



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Dissect the question: "where is the best burger place in the Philippines?"

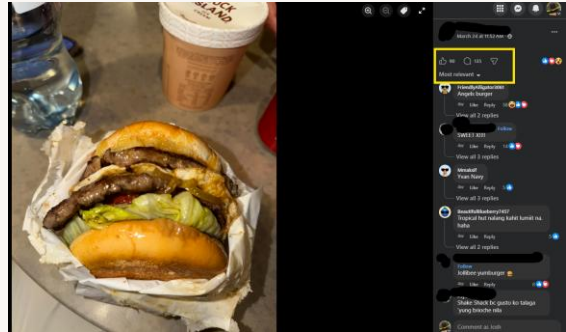
- What does "best" mean? Could be customer reviews and net promoter score ("I highly recommend x").
- Do we include all burger stores in the Philippines?

Defining data analysis and the data analysis process

III. Data: Identifying data sources >>

This step encompasses activities related to identifying the data required and available, ensuring its usability for solving analytics problems; and fulfilling documentation and reporting needs.

Ex. Facebook comments



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Mention data management plan

III. Data: Collect and profile data >>

Ex. Manual extraction of Facebook comments to Excel

					An at 2026-03-24 03:25 PM	
					https://www.facebook.com/groups/futaba-in-yaher/permalink/1201025359348671/?	
1	Share	- Comment	< Reactions >	< Replies >		
2		👍 Love Burger, Mela Sa. Makati		7	3	
3		Angels Burger		38	2	
4		Kungis Thomas				
5		It cuth for the win				
6		Pigapple				
7		Shake Shack bu gusto ko talaga yang british nla		5		
8		Sweet X		1		
9		Sweet X		1		
10		Sweet X		1		
11		Sweet X or Greenlight Burgers		1		
12		King's Premium Burger 🍔 🍷 🥂 Bis Cuz Laguna				
13		Infinite Yum Burger for the win 🏆		1		
14		Where Burger 🍔				
15		infinite yumburger 🍔		6		
16		TROPICAL HOT HAMBURGERS				
17		ZBurga Bawang pa rin 🍔 🍷				
18		Sweet X		1		
19		Messina Burger				
20		Bun Run				
21		Tales in Publication!!!!				
22		Our family would enjoy burgers once in a while but my brother's a something we look forward to, it's just out of this world.				
23		Army Nang pa rin				
24		Shake Shack sa Maginhawa, QC ☺️				
25		Ven Venn		2	2	
26		wahahhhhh na UP 🍔 🍷 🍷 🍷				
27		I'm interested to try Incheon..			1	
28		Sweet off!		1		
29		Infinite Yum burger 🍔				

Mention data dictionary

Defining data analysis and the data analysis process

III. Data: Data cleansing



After collecting the relevant data, data cleansing and standardization is needed to ensure that we can perform accurate analysis.

Ex. Standardization of shops, exploding responses

ID	User	Comment	Responsible Shop	Shop	Remarks
1	Jayvee Montebello	SWEET X!!!	7	Sweet X Eatery	
2	AnonymousUser1	Love Burgers, Mula St. Mula		Love Burgers	
3	Friend4Alligator0881	Angels burger	30	Angel's Burger	
4	Jay Dee	Hungry Thirsties		Hungry Thirsties	
5	ShaneKingfisher	It cuts for the win	14	Shane Kingfisher	
6	ProduceMarket0000000000	Plastic		Produce Market	
7	Psyche Cando	Plastic Shuck for gusto ko talaga 'yang binuho nila	5	Psyche Cando	
8	Clare Dato	Sweet X	1	Sweet X Eatery	
9	Silvestre040796	Sweet X	1	Sweet X Eatery	
10	El Francisco Antonio	Sweet X	1	Sweet X Eatery	
11	Patricia Henry de Dios	Sweet X or Greenlight Burgers	1	Sweet X Eatery	
12	Denielle Anna Villegas	King's Premium Burgers 🍔 🍷 🍷 🍷 The Great Laguna	1	King's Premium Burgers	
13	Anting0285	Juliane Tum Burger for the win 🍔 🍷	1	Juliane	
14	Jason Yu Aranga	Wham Burger 🍔		Wham Burger	
15	Zhuangmei Sun Juan Santos - Santos	Juliane burger 🍔		Juliane	
16	Raul Lemery	THEYCA, NOT FARMHOUSE	6	THEYCA	
17	LA Puyangas Ocadin	2Bugs Bazaar pa rin 🍔 🍷		2Bugs Bazaar	
18	Choi Epithio	Sweet X	1	Sweet X Eatery	
19	Bella Vison Alud	Mula Burger		Mula Burger	
20	Norree Norree	Bun Bun		Bun Bun	
21	Determiend040304	Take in Publication!!!!		KA MNL	
22	Chella G	Our family loved this burger since it's a while but my brother is so something we both loved it, it's just out of this world			Shop not specified
23	Andreabagpoc1158	Amey Hery pa rin		Amey Hery	
24	Derek Dorelva	Plastic Shuck sa Maghihawa, QC! 🍔		Plastic Shuck	
25	Mrsakul	Yuan King	2	Yuan King	
26	Paul Juliano	crunchy shuck sa UP 🍔 🍷 🍷		UP Plastic Shuck	
27	James Jake	I'm interested to try Sautee...		Sautee	
28	Wagdy	Sweet X!!	1	Sweet X Eatery	
29	Andrea Nicole Elorite Salas	Juliane Tum burger 🍔		Juliane	
30	Nighting Knight	Burger machine		Burger Machine	
31	Ingolf718	1875 Old Fashioned Burgers	1	1875 Old Fashioned Burgers	
32	Isay Goley	Sweet X 🍔	1	Sweet X Eatery	
33	Sejap Lantoran	Bun Burger		Bun Burger	

Defining data analysis and the data analysis process

IV. Methodology selection



This involves selecting the appropriate analytical methods, tools, and technologies to solve the defined analytics problem.

Ex. Simple tabulation of the responses

Cleaned Names	Count of Id	Sum of Reactions	Sum of Replies
Sweet X Ecstasy	13	16	3
8Cuts	4	14	
Jollibee	4	7	
Raging Bull Burger	3		
The Snack Shack	3		
Shake Shack	3	5	
Burger Machine	3		
Tropical Hut	3	4	2
Flipside Burgers	3	1	
1975 Old-Fashioned Burgers	3	1	
Love Burgers	2	1	
Hungry Homies	2	1	
Bun Run	2	1	
ArmyNavy	2		
Zark's Burger	2		
Sautéo	2		1
Scratch Smash Burgers	2	2	
Angel's Burger	2	39	2
Eat Sleep Burger	1		
Wham! Burgers	1		
Teddy's Bigger Burgers	1		
HEY HO Garage Burgers	1		

Defining data analysis and the data analysis process

V. Analytics / model development>>

This covers activities related to designing, building, and refining models, as well as documenting and communicating the results.

Ex. Visualized result of the responses based on how frequent the shop was mentioned.

Is Sweet X Ecstasy the best burger place?
Burger places ranked by comments.



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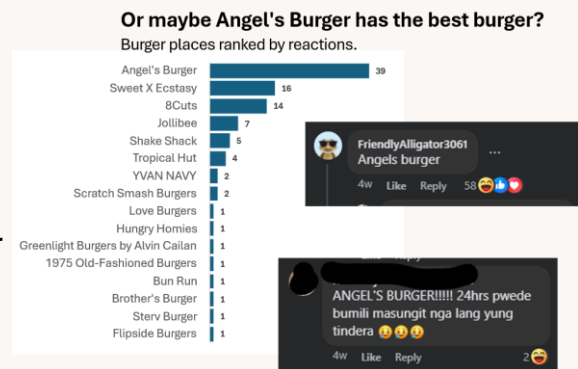
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Defining data analysis and the data analysis process

V. Analytics / model development>>

Alternative analysis and model build can also be used for comparison.

Ex. Visualized result of the responses based on the number of reactions on the comments.



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Understanding the concept of the data matters here.

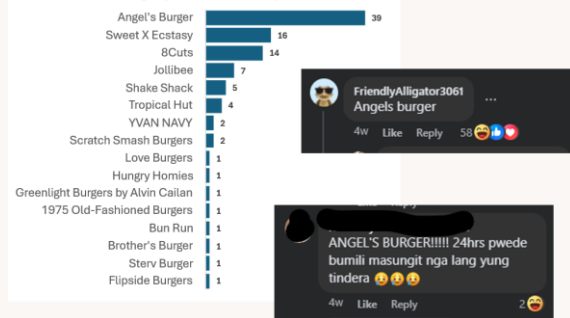
VI. Deployment



This covers activities related to delivering the analytics solution and ensuring it is effectively integrated into the business environment.

Ex. Can we automate data collection and build a dashboard?

Or maybe Angel's Burger has the best burger?
Burger places ranked by reactions.



VI. AS Lifecycle management



This involves ongoing oversight and calibration to ensure the analytics solution continues to perform effectively and delivers valid results over time that continue to deliver business value.

Ex. How often do we rerun the analysis?

Or maybe Angel's Burger has the best burger?
Burger places ranked by reactions.



Defining data analysis and the data analysis process

The 4 types of analytics



- Descriptive: "What happened?"
- Diagnostic: "Why did it happen?"
- Predictive: "What will most likely happen?"
- Prescriptive: "How should we take action?"

Exploratory data analysis (EDA)

Learn the art and science of doing exploratory data analysis

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Exploratory data analysis (EDA)

Exploratory data analysis defined >>

Exploratory data analysis (EDA) is an open-ended, iterative data analysis method designed to unearth patterns, anomalies, relationships, or insights without preconceived notions.

Confirmatory data analysis (CDA) tests pre-defined hypotheses, evaluate evidence, and challenge or validate theoretical models using exact significance, inference, and confidence estimates.

Exploratory data analysis (EDA)

Exploratory data analysis defined >>

CDA: "Test if the training helped our agents get better CSAT performance."

EDA: "Check what other factors that could have led to higher or lower performance on CSAT."

Exploratory data analysis (EDA)

Four principles of EDA (4R's) >>

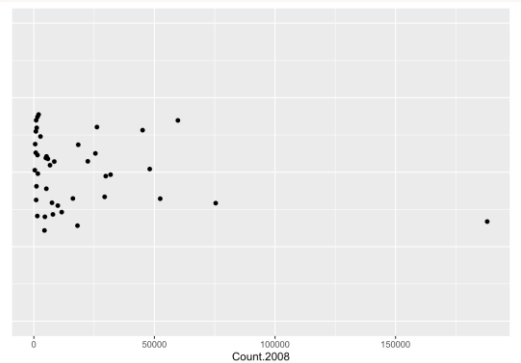
- Revelation: Emphasis on using graphs to find patterns or displaying fits
- Resistance: Describe the general pattern in the majority of the data
- Reexpression: Rescale the data for standardization
- Residual: FIT means general pattern. Residuals = data - FIT

Exploratory data analysis (EDA)

Ex. 4R's: revelation & resistance >>

2008 US immigrants by country of origin

- Most immigration counts are clustered between 0-30k
- Data is right-skewed with Mexico having >188k immigrants



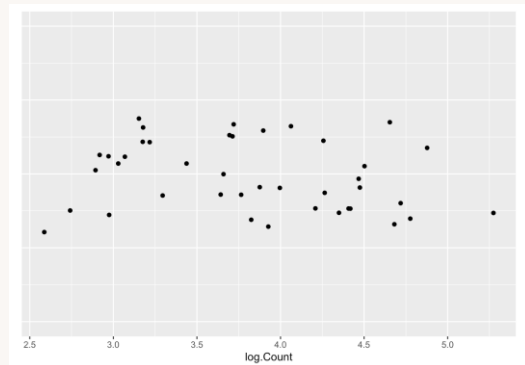
Exploratory data analysis (EDA)

Ex. 4R's: reexpression



2008 US immigrants by country of origin (log10)

- Skewness was eliminated
- Easier to detect "clumping" of countries



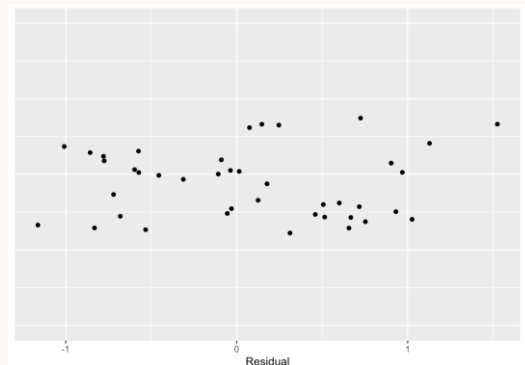
Exploratory data analysis (EDA)

Ex. 4R's: residual



2008 US immigrants by country of origin (log10 - FIT)

- FIT = 3.75
- 2 countries are 1 smaller than average
- 1 country is ~1.5k larger than ave



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Let's return to our questions about these data to see what we have learned in this brief data analysis:

What countries have contributed the most immigrants to the United States?

ANSWER: Mexico is by far the leader in supplying immigrants to the U.S.

What is a typical number of immigrants from a country?

ANSWER: On the log scale, a typical immigrant count is 3.75.

Are Asian countries contributing more or less immigrants than countries from Europe?

ANSWER: We didn't address this question in our brief analysis, but we'll soon talk about how we can compare batches of data.

The magazine *Time* had an issue that focused on the U.S./Mexico border. Are we getting an unusually large number of immigrants from Mexico?

ANSWER: Yes – this analysis confirms that Mexico is supplying many immigrants.

Which countries are contributing a large number of immigrants relative to their population size?

ANSWER: In order to answer this question, we would need to collect the populations of the countries in the table. This would be an interesting study.

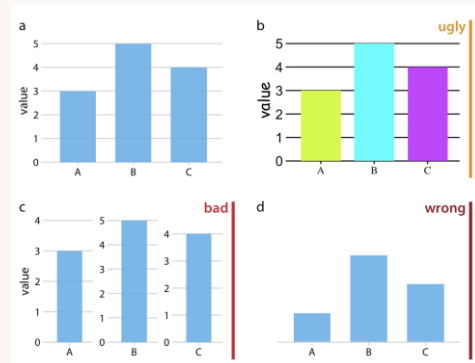
Fundamentals of data visualization

A primer on making informative and compelling figures

Ugly, bad, and wrong figures



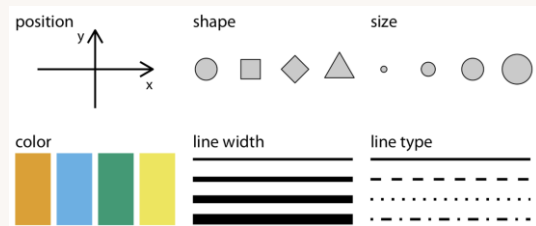
- a. Reasonable viz with no major flaw
- b. Technically correct, but not aesthetically pleasing
- c. Misleading due to different scales
- d. Values cannot be ascertained



Commonly used aesthetics



Some of these aesthetics can represent both continuous and discrete data (position, size, line width, color), while others can usually only represent discrete data (shape, line type).



Types of variables



- a. Quantitative: numerical (1.3, 5.7, 83, 1.5×10^{-2}) or discrete (1, 2, 3)
- b. Qualitative: unordered (dog, cat, fish) or ordered (good, fair, poor)
- c. Date or time: Jan. 5 2018, 8:03am
- d. Text: The quick brown fox jumps over the lazy dog.

Fundamentals of data visualization

Ex. Mapping data onto aesthetics >>

Table 2-2. First 8 rows of a dataset listing daily temperature normals for four weather stations. Data source: National Oceanic and Atmospheric Administration (NOAA).

Month	Day	Location	Station ID	Temperature (°F)
Jan	1	Chicago	USW00014819	25.6
Jan	1	San Diego	USW00093107	55.2
Jan	1	Houston	USW00012918	53.9
Jan	1	Death Valley	USC00042319	51.0
Jan	2	Chicago	USW00014819	25.5
Jan	2	San Diego	USW00093107	55.3
Jan	2	Houston	USW00012918	53.8
Jan	2	Death Valley	USC00042319	51.2

Fundamentals of data visualization

Ex. Mapping data onto aesthetics >>

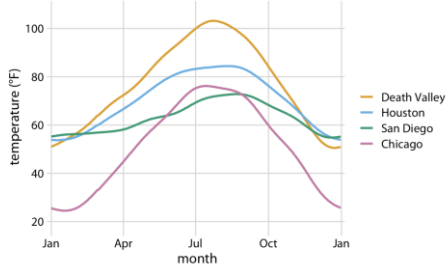


Figure 2-3. Daily temperature normals for four selected locations in the US. Temperature is mapped to the y axis, day of the year to the x axis, and location to line color. Data source: NOAA.

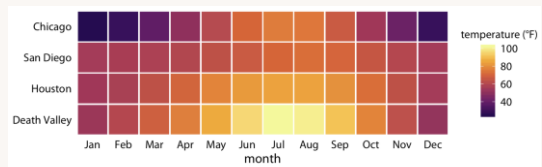


Figure 2-4. Monthly normal mean temperatures for four locations in the US. Data source: NOAA.

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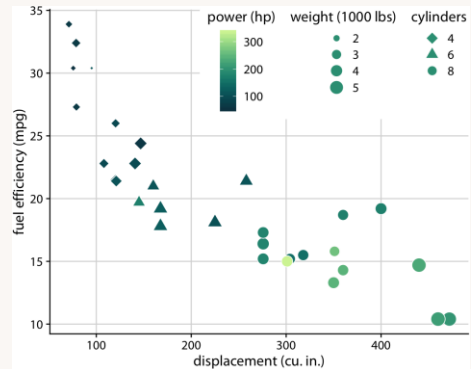
Poll: what graph do you like the most?

Fundamentals of data visualization

Ex. Mapping data onto aesthetics >>

```
> mt.cars
```

	mpg	cyl	displ	hp	drct	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
Duster 360	14.3	8	360.0	245	3.21	3.570	15.44	0	0	3	4
Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2
Merc 230	22.8	4	140.8	95	3.62	3.150	22.90	1	0	4	2
Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4
Merc 280C	17.8	6	167.6	123	3.92	3.440	18.90	1	0	4	4
Merc 450SE	16.4	8	275.8	180	3.07	4.070	17.40	0	0	3	3
Merc 450SL	17.3	8	275.8	180	3.07	3.730	17.60	0	0	3	3
Merc 450SLC	15.2	8	275.8	180	3.07	3.780	18.00	0	0	3	3
Cadillac Fleetwood	10.4	8	472.0	205	2.93	5.250	17.98	0	0	3	4
Lincoln Continental	10.4	8	460.0	215	3.00	5.424	17.82	0	0	3	4
Chrysler Imperial	14.7	8	440.0	230	3.23	5.345	17.42	0	0	3	4
Fiat 128	32.4	4	78.7	66	4.08	2.200	19.47	1	1	4	1
Honda Civic	30.4	4	75.7	52	4.93	1.615	18.32	1	1	4	2
Toyota Corolla	33.9	4	71.1	65	4.22	1.835	19.90	1	1	4	1
Toyota Corona	21.5	4	120.1	97	3.70	2.465	20.01	1	0	3	1
Dodge Challenger	15.5	8	318.0	150	2.76	3.520	16.47	0	0	3	2
AMC Javelin	15.2	8	304.0	150	3.15	3.435	17.30	0	0	3	2
Camaro Z28	13.3	8	350.0	245	3.73	3.840	15.41	0	0	3	4
Pontiac Firebird	19.2	8	400.0	175	3.08	3.845	17.05	0	0	3	2
Fiat X1-9	27.3	4	79.0	66	4.08	1.935	18.90	1	1	4	1
Porsche 914-2	26.0	4	120.3	91	4.43	2.140	16.70	0	1	5	2
Lotus Europa	30.4	4	95.1	113	3.77	1.513	16.90	1	1	5	2
Ford Pantera L	15.8	8	351.0	264	4.22	3.170	14.50	0	1	5	4
Ferrari Dino	19.7	6	345.0	175	3.62	2.770	15.50	0	1	5	6
Maserati Bora	15.0	8	301.0	335	3.54	3.570	14.60	0	1	5	8
Volvo 142E	21.4	4	121.0	109	4.11	2.780	18.60	1	1	4	2



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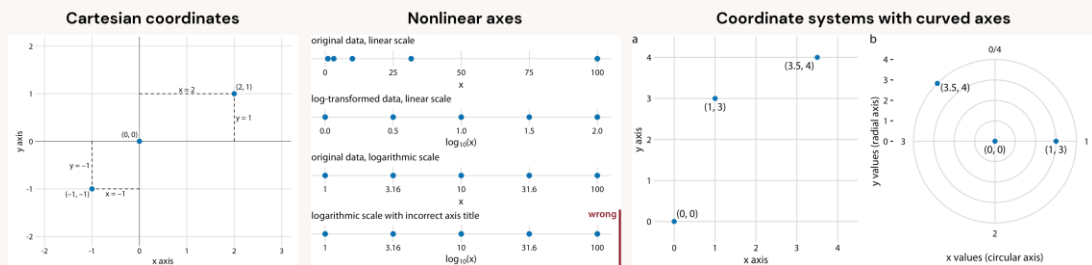
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Figure 2-5 uses five scales—two position scales, one color scale, one size scale, and one shape scale—and each scale represents a different variable from the dataset.

Figure 2-5. Fuel efficiency versus displacement, for 32 cars (1973–74 models). This figure uses five separate scales to represent data: (i) the x axis (displacement); (ii) the y axis (fuel efficiency); (iii) the color of the data points (power); (iv) the size of the data points (weight); and (v) the shape of the data points (number of cylinders). Four of the five variables displayed (displacement, fuel efficiency, power, and weight) are numerical continuous. The remaining one (number of cylinders) can be considered to be either numerical discrete or qualitative ordered. Data source: Motor Trend, 1974.

Coordinate system and axes



Fundamentals of data visualization

Coordinate system and axes



Chart using cartesian coordinate

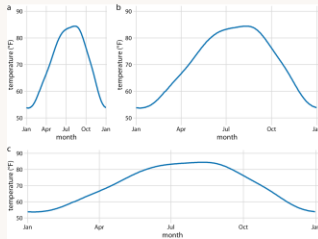


Chart with scaled values

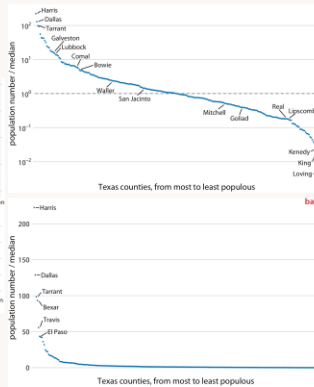
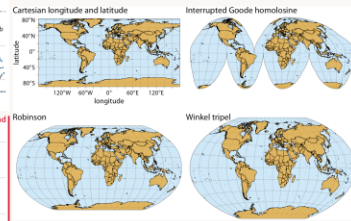


Chart using the polar coordinate system



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Color scales

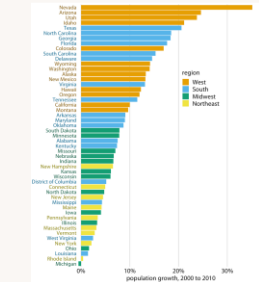


There are three fundamental use cases for color in data visualizations.

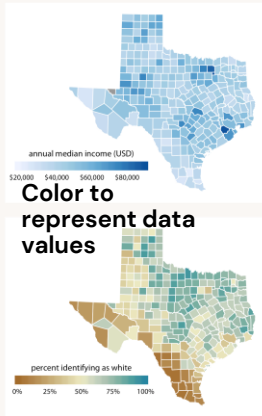
- Color as a tool to distinguish
- Color to represent data values
- Color as a tool to highlight

Fundamentals of data visualization

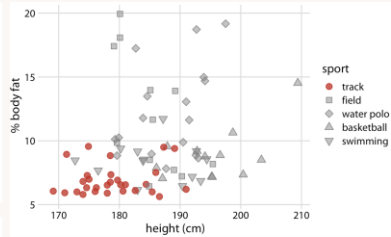
Color scales



Color as a tool
to distinguish



Color to
represent data
values



Color as a tool to
highlight

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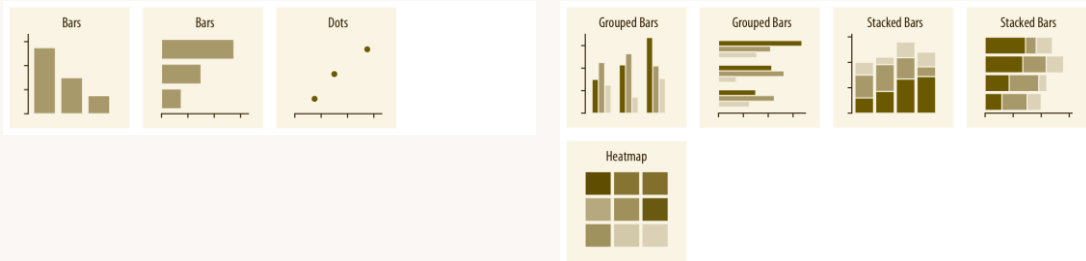
Directory of visualizations



Here are the categories of visualizations based on what they show.

- Amounts
- Distributions
- Proportions
- X-y relationships
- Geospatial data
- Uncertainty

Charts for showing amounts >>



Fundamentals of data visualization

Charts for showing distributions >>

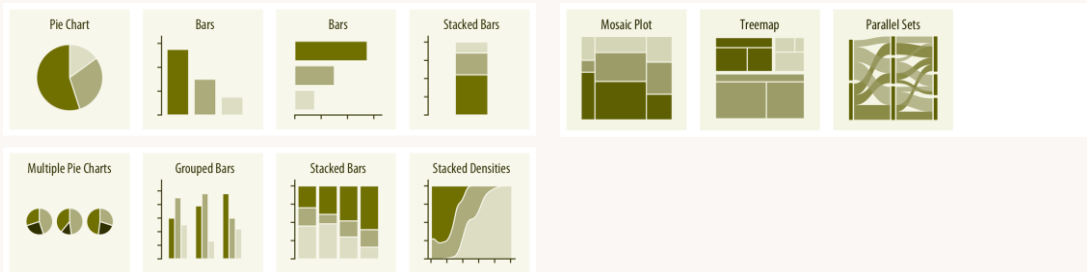


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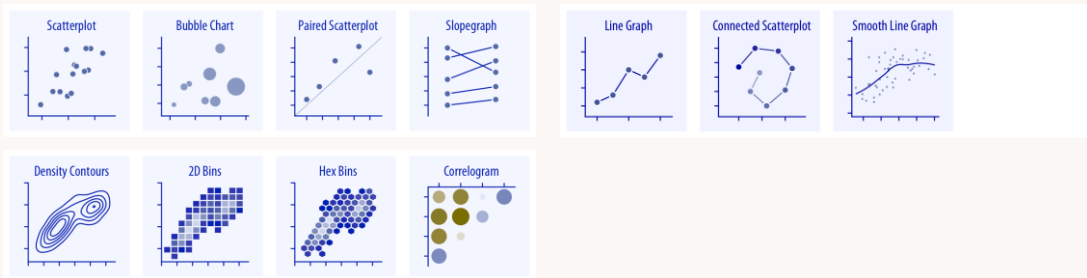
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Charts for showing proportions >>



Fundamentals of data visualization

Charts for x-y relationships

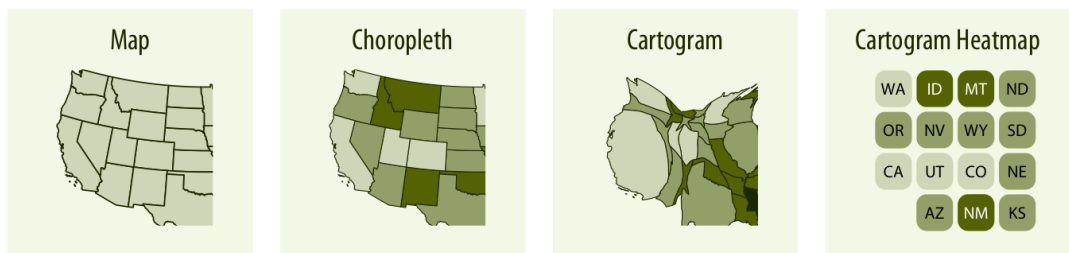


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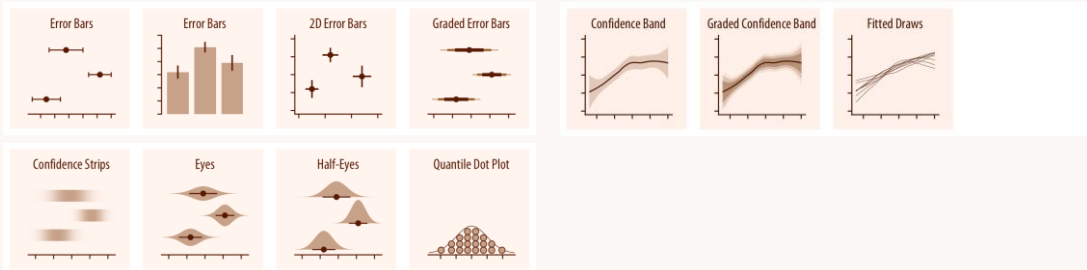
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Charts for geospatial data



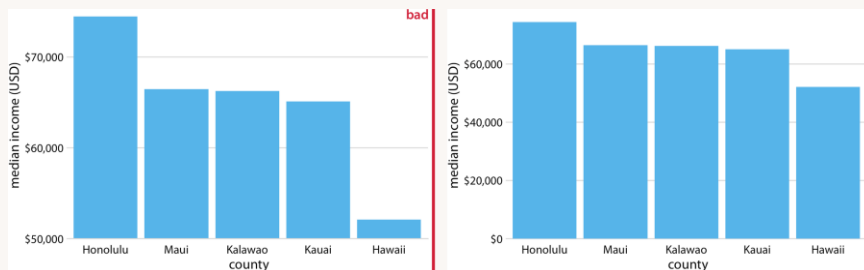
Charts for showing uncertainty >>



Principles of figure design



The principle of proportional ink: The sizes of shaded areas in a visualization need to be proportional to the data values they represent.



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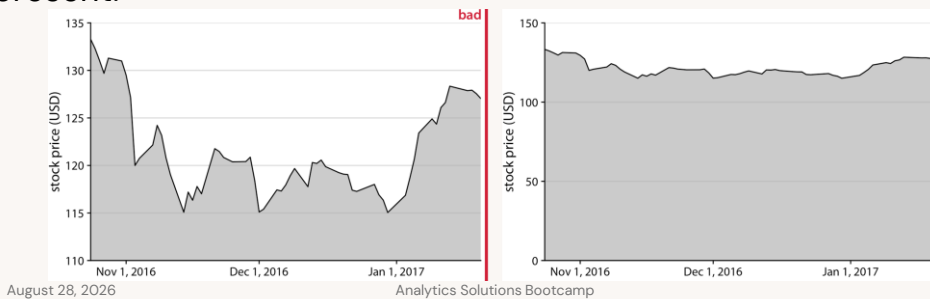
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Principles of figure design



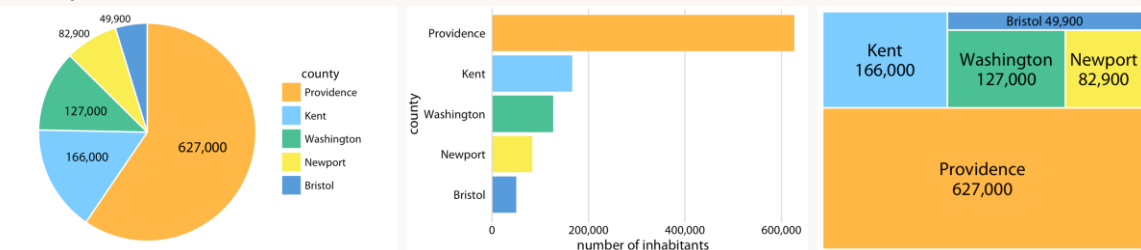
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Principles of figure design



The principle of proportional ink: The sizes of shaded areas in a visualization need to be proportional to the data values they represent.



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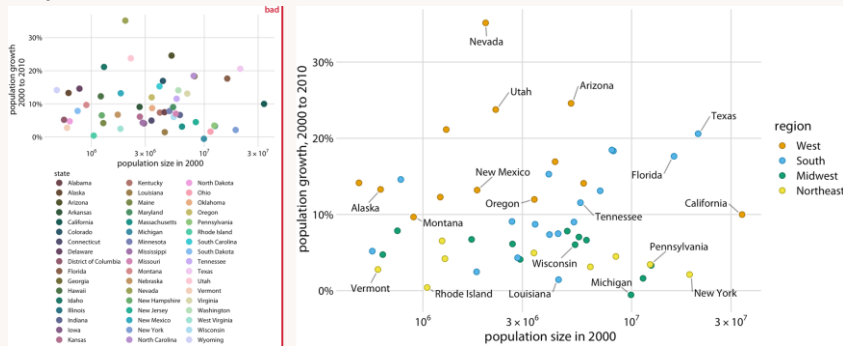
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Principles of figure design



Common pitfalls of color use: too much information



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Principles of figure design



Common pitfalls of color use: not designing for color-vision deficiency



Descriptive analytics

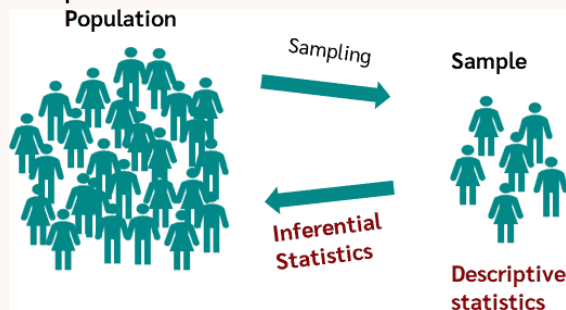
Data analysis to answer the question “what happened?”

Descriptive analytics

Descriptive analytics



Descriptive analytics uses statistical summaries and data visualization techniques to condense and describe historical data.



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Descriptive analytics

Descriptive analytics



Descriptive analytics uses statistical summaries and data visualization techniques to condense and describe historical data.

Descriptive analytics can be classified as follows:

- Univariate – for describing distribution of a single variable
- Bivariate – for describing the relationship of two variables
- Multivariate – for analyzing three or more variables

Descriptive analytics

Univariate analysis



Examines one single variable to determine pattern, spread, and outliers.

- Measures of central tendency: mean, median, mode
- Measures of dispersion: range, IQR, variance and standard deviation
- Other measures: skewness and kurtosis

Bivariate analysis



Bivariate analysis explores the strength of relationship of two variables, or if the differences between the two variables are significant.

- Numerical & numerical: height vs weight, study hours vs test scores
- Categorical & categorical: gender vs product preferences
- Categorical & numerical: education level vs income

Descriptive analytics

Hypothesis testing



A statistical method used to determine whether there is sufficient evidence to support a claim about a population.

Hypothesis testing concepts >>

- Null Hypothesis (H_0) – Assumes no significant difference, relationship, or effect.
- Alternative Hypothesis (H_1) – Represents the difference, relationship, or effect we are testing for.
- Significance Level (α) – The threshold for deciding whether the evidence is statistically significant, commonly 0.05.
- p-value – Indicates how likely the observed result would be if the null hypothesis were true.
 - $p\text{-value} \leq \alpha \rightarrow \text{Reject } H_0$
 - $p\text{-value} > \alpha \rightarrow \text{Fail to reject } H_0$

Bivariate analysis



Statistical tests performed on two variables.

- Correlation analysis: measures the strength and direction of relationship of 2 variables
- Chi-square test: Tests whether there is a significant relationship between categorical variables using frequencies or counts.
- T-tests: used to Compares the means of two groups to determine whether they are significantly different.

Multivariate analysis



Statistical tests performed on three or more variables.

- Regression analysis: analyzes the relationship between one or more independent variables and one dependent variable.
- Principal component analysis (PCA): dimensionality reduction technique and helps us to reduce the number of features in a dataset while keeping the most important information.
- Clustering: used to identify natural groupings or clusters within a dataset

Diagnostic analytics

Data analysis to answer the question “what happened?”

Diagnostic analytics

Diagnostic analytics



Diagnostic analytics analyzes historical datasets to uncover root causes, patterns and relationships. It uses techniques such as data mining, correlation analysis, drill-down methods and statistical modeling.

Diagnostic analytics techniques >>

- Drill down analysis: breaking down metrics by different dimensions
- Root cause analysis (RCA): 5 why's, Ishikawa (fishbone) diagram
- Pareto analysis: 80/20 rule states that roughly 80% of effects come from 20% of causes
- Statistical techniques such as regression analysis, hypothesis testing, cluster analysis, time-series decomposition

Business intelligence (BI) primer

Data analysis to answer the question “what happened?”

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Business intelligence



Business Intelligence (BI) is the process of turning organizational data into information that supports better business decisions.

- More than just a report builder, BI should answer business questions

Your role as BI developer / analyst >>

- Create reports / dashboards that addresses user needs
- Optimizing reports may mean automating their manual data collection processes
- Empower user on leveraging data for business decisions

Semantic model / analytics layer >>

Semantic table / model refers to the design of the tables you will use for building the dashboards and reports.

- Flattened version
- Star / snowflake schema

Business intelligence primer

BI stack

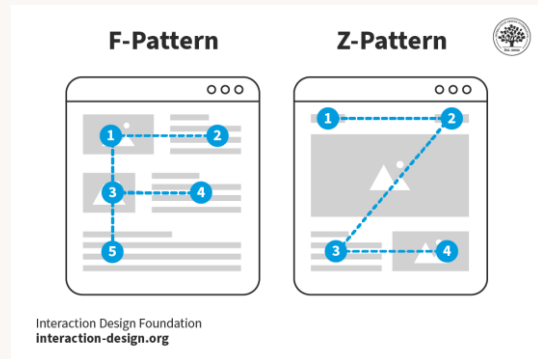


- Excel
- SQL
- Power BI
 - Power Query
- Tableau
 - Tableau Prep Builder

Dashboard design principles



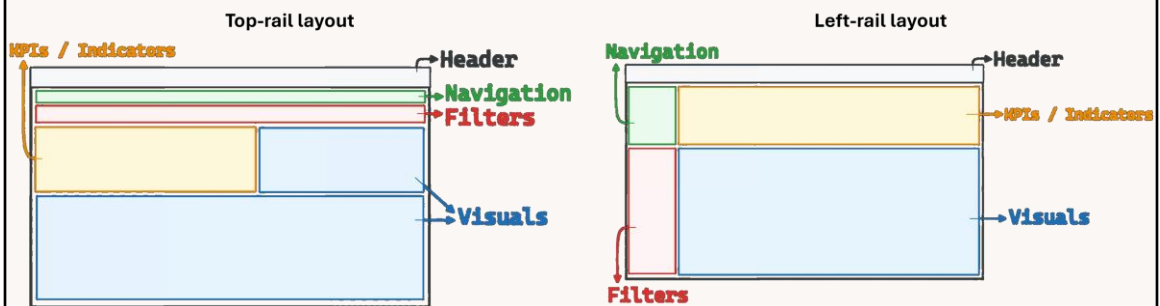
Use F or Z pattern. This refers to the pattern when humans reads.



Dashboard design principles



Common dashboard layouts



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Dashboard design principles



Identify what type of dashboard is needed and what kind of users will use the dashboard.

Dashboard Type	Purpose	User	Cadence	Design Priorities	Example
Analytical	Root cause analysis	Analysts	Ad-hoc / Deep dive	High interaction, filters, drill-downs	Sales deep-dive
Operational	Live monitoring	Shift Leads	Real-time	Low latency, big status, alerts	Support wallboard
Strategic	Long-term steering	Executives	Quarterly / Monthly	Comparisons, baselines, annotations	KPI summary
Tactical	Daily execution	Managers	Daily / Weekly	Actionability, progress vs. targets	Campaign tracker
Explanatory	Storytelling	General Audience	As-needed	Narrative, minimal controls	Broad overview

No *One* Plays On *One* Foot.

Explore how one click transforms into customers, belonging, and shared identity. **Grounded in human psychology**, we reveal why these moments ignite revenue growth and spikes.

[Take the First Step →](#)

[▶ Watch the Play](#)

 Photos by: Mateo Franciosi

 Report by: Jerome Jan Arbonida



It Starts with One Step

One click turns to traffic. Traffic becomes customers. Customers drive revenue.



Total Traffic
8,832,694



Conversion Rate **3.24%**
Revenue per 100 Traffic **€ 128.79**

Total Customers
28,876



YoY Customers Growth **4.41%**
Revenue per 100 Customers **€ 39,393.96**

Total Orders
101,681



YoY Orders Growth **29.59%**
Revenue per 100 Orders **€ 11,187.34**

Total Revenue
€ 11,375,399.67



YoY Revenue Growth **30.09%**
Primary Revenue Driver **Customers**

Pick a driver to see how it plays with revenue.

Total Traffic

Total Customers

Total Orders

Pick a year to focus the sales game.

2021

2022

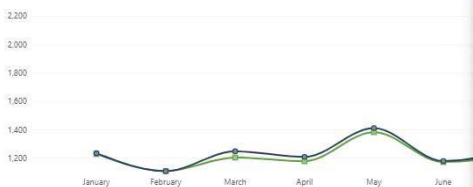
2023

2024

2025

Revenue is typically strongest in August, during Kit Launch Season, making it a key period for overall sales performance.
Customer levels peak in December, during Holiday Peak, highlighting the period of highest customer engagement and demand across 2021-2025.

— Hover to reveal 5-year trends



Why This Behavior?

RECOMMENDATIONS

- Activate customers through the **'No One Plays On One Foot'** (no one buys alone) campaign.
- Design **moments where buying feels shared**, visible, and socially reinforced, so one action naturally triggers another.
- **Explore the next page** to see how customer dynamics across countries amplify this behavior into collective growth.



What Should We Do Next?





We Become our Own Team

One symbol—shaping a shared identity that forms our own winning team.

1 2 3 4

Total Orders

101,681

YoY Orders Growth 29.59%



Total Revenue

€ 11,375,399.67

YoY Revenue Growth 30.09%



Order-Revenue Insights

Across products, stronger order volume is closely tied to revenue performance: a change of 472 orders typically corresponds to about €67,349 in revenue.



Velocity 11 Elite Footwear from AeroKick leads demand with 2,293 orders, the highest among selected products.

Pick a driver to see how it plays with revenue.

Total Orders YoY Orders Growth Revenue per 100 Orders Total Revenue

Who Drives the Orders

Casual Fan Hardcore Fan Collector



Where Orders Happen

Desktop Web iOS and Android App



What Customers Order Most

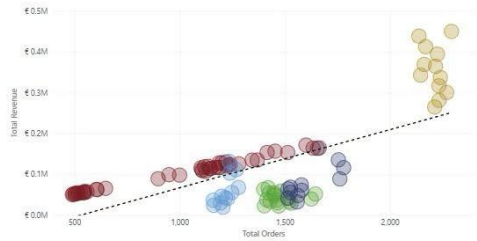
Kits and footwear drive orders and revenue.



Why This Behavior?

Pick a year to focus the sales game.

2021 2022 2023 2024 2025



What Should We Do Next?



References

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